Palindrome Check

AlgoExpert · Easy · Strings

The problem

- Does the string read the same forwards and backwards?

Intuition

- Compare characters from both ends moving inward; any mismatch means it isn't a palindrome.

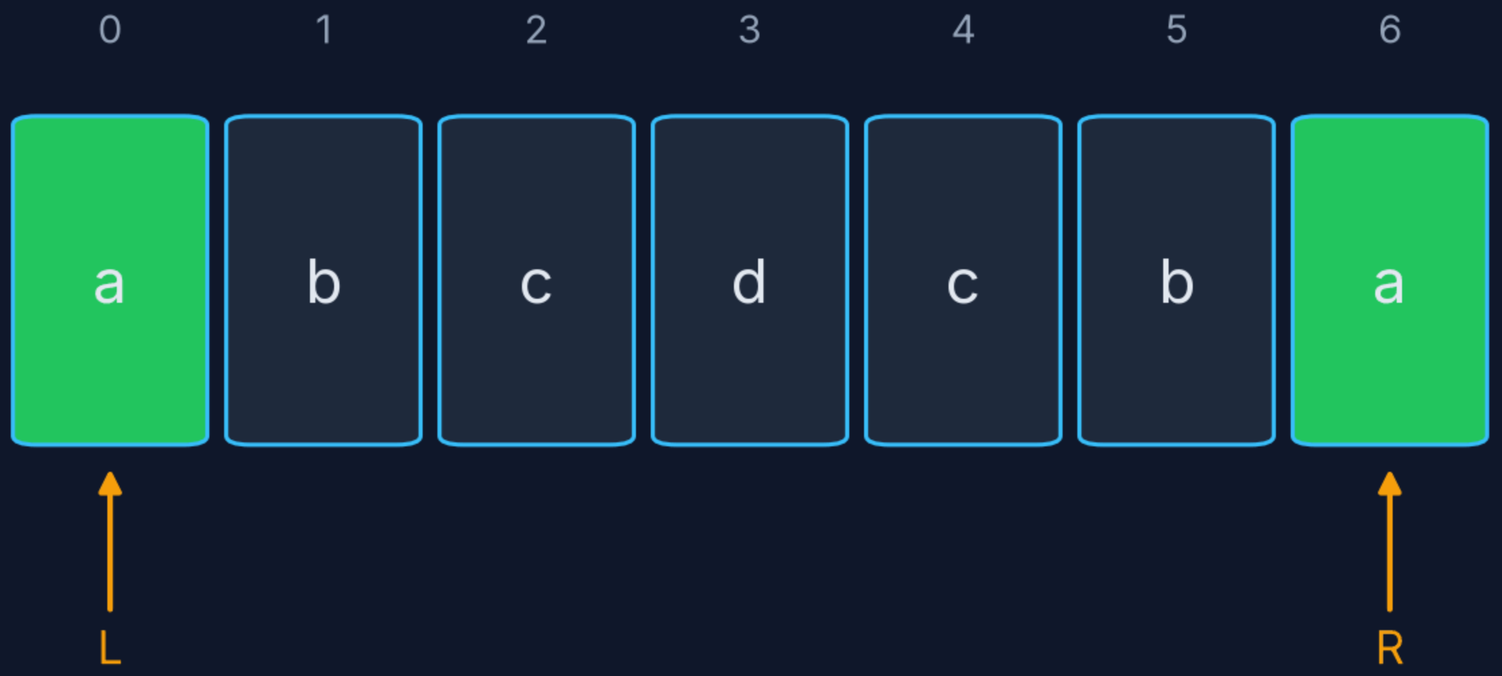
Brute force vs optimal

- Reverse the string and compare → O(n) time, O(n) space
- Naive concatenation in a loop → $O(n^2)$
- Two pointers → O(n) time, O(1) space

Approach

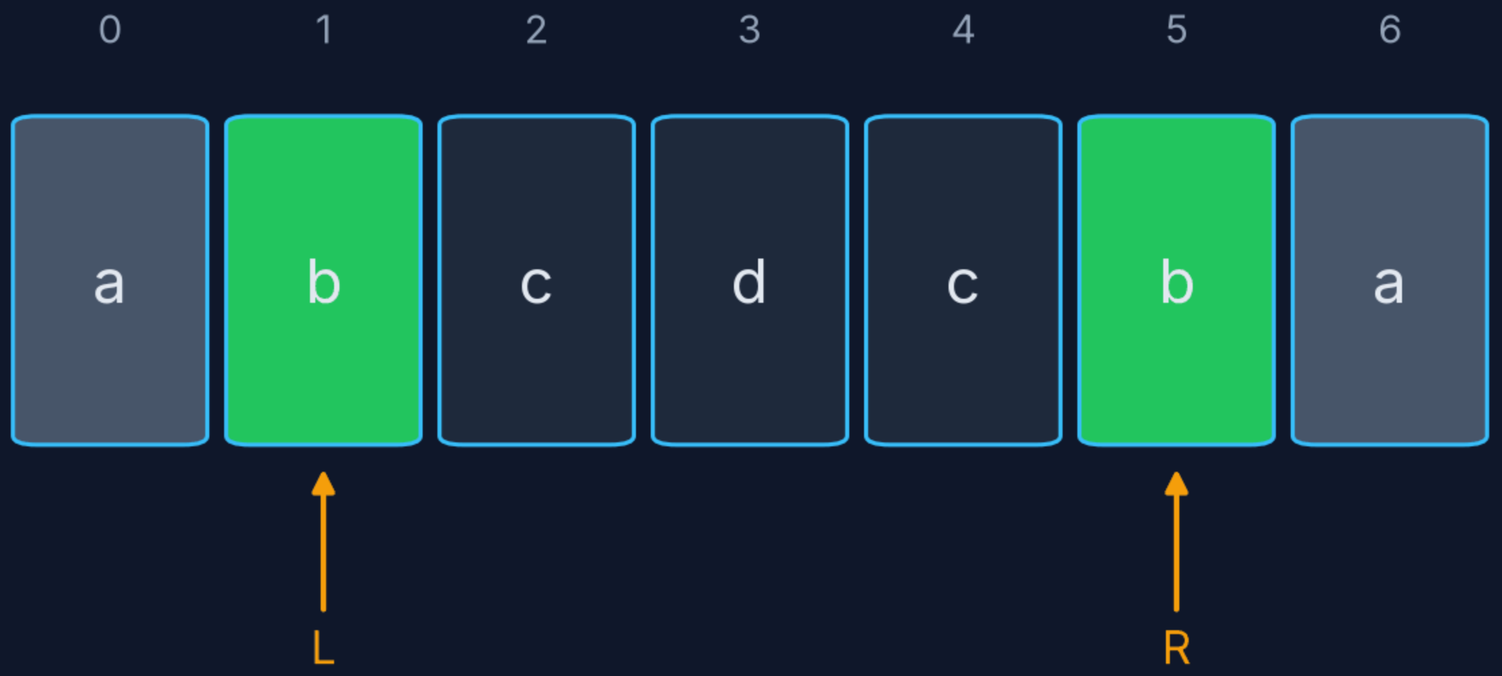
- 1. Left pointer at the start, right at the end
- 2. Compare and move inward
- 3. Stop when they meet

Worked example



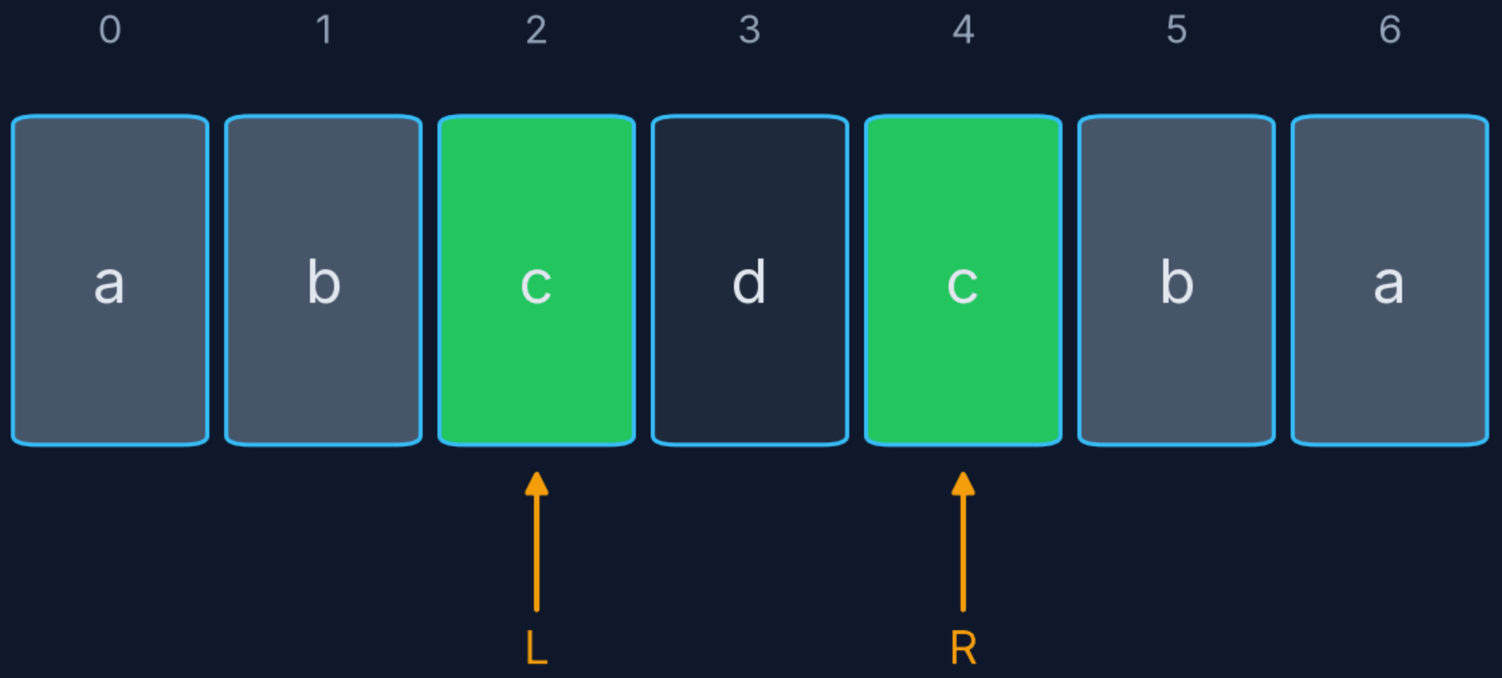
'a' = 'a' → move inward

Worked example



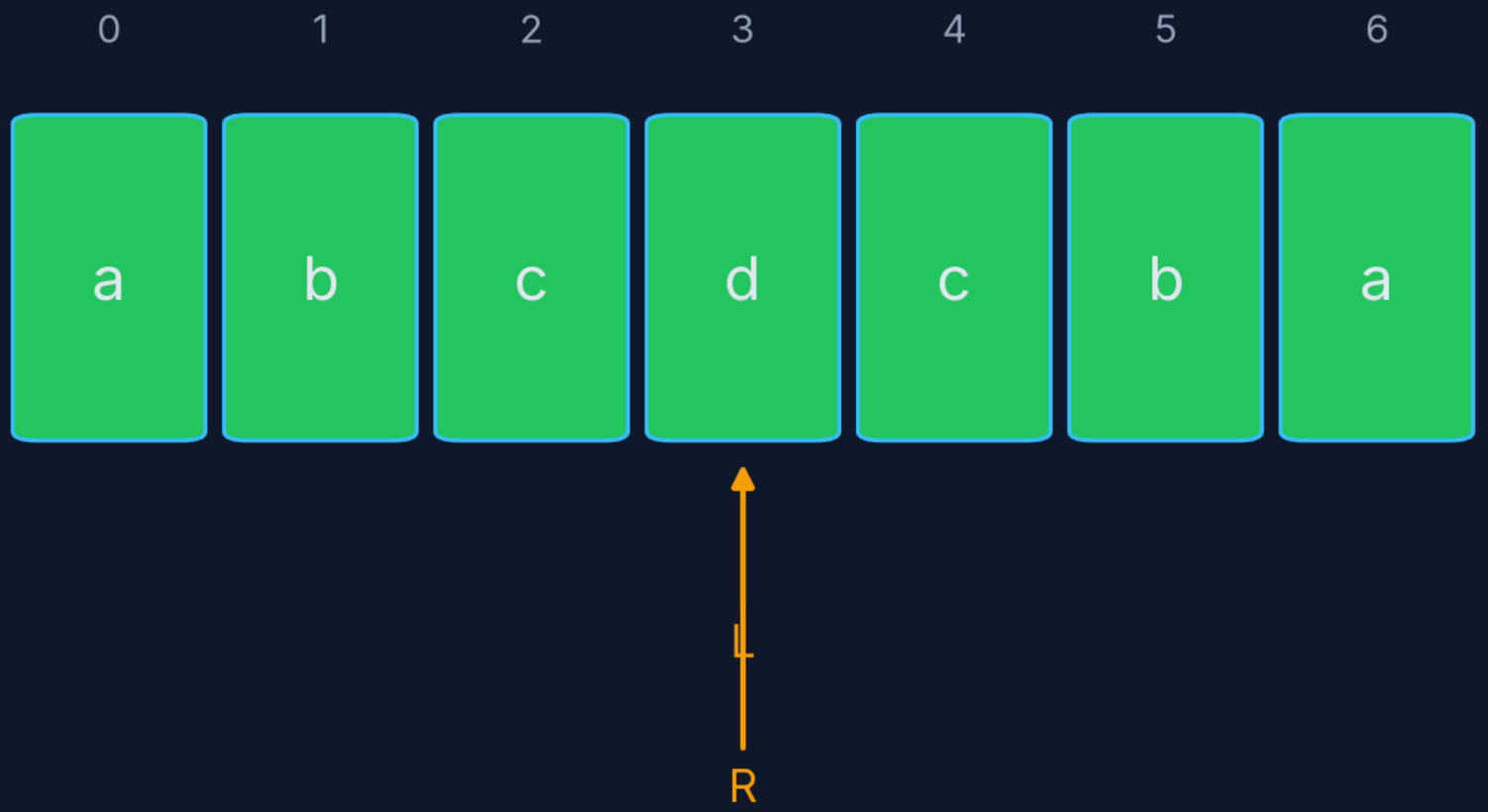
'b' = 'b' → move inward

Worked example



'c' = 'c' → move inward

Worked example



Pointers met → palindrome

Java solution

```java
public class PalindromeCheck {
    public static boolean isPalindrome(String s) {
        for (int i = 0, j = s.length() - 1; i < j; i++, j--) if (s.charAt(i) != s.charAt(j)) return false;
        return true;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Edge cases & pitfalls

- Empty / single char → true
- Even vs odd length
- Clarify case and non-letters

Interview tips & follow-ups

- Follow-up: valid palindrome with one deletion
- Longest palindromic substring

Recap

- Compare characters from both ends moving inward; any mismatch means it isn't a palindrome.
- Complexity: O(n)